Stage1: Open Applications

Students need to submit a detailed proposal in PDF format that includes:

1. **Title of the project / product**
   VR-Based Immersive Motor Rehabilitation System with Custom Haptic Controller for Physiotherapy
2. **Problem Statement:**
   Conventional physiotherapy depends on repetitive exercises that often reduce patient motivation and therapy adherence, slowing recovery. Progress is mainly assessed through observation, with limited objective data on movement, grip, force, or range of motion. Advanced rehabilitation systems are expensive and inaccessible for many hospitals and clinics. There is need for an affordable, engaging, data-driven rehabilitation solution that improves participation and supports better clinical outcomes.
3. **Solution proposed:**
   The proposed solution is a VR-Based Immersive Motor Rehabilitation System with Custom Haptic Controller designed for upper-limb physiotherapy. It combines a custom-built controller with motion and force sensing, microcontroller-based data acquisition, data processing and signal mapping, and an immersive Virtual Reality environment. The controller captures the patient's hand/arm movement, grip, applied force, and Range of Motion (ROM), and maps these inputs to interactive VR-based rehabilitation exercises. Real-time visual and audio feedback keeps the patient engaged, while a therapist dashboard records performance data for progress tracking and treatment monitoring. The system uses gamification, progressive exercises, and performance analytics to reduce exercise monotony and support personalized rehabilitation.

4. **Implementation Plan:**
   The solution will be implemented in a phased manner, beginning with clinical requirement analysis, literature study, product specifications, ergonomic CAD design, and development of the custom rehabilitation controller. The next phase includes sensor, PCB, microcontroller and electronic integration, followed by prototype fabrication. VR-based rehabilitation activities, motion tracking, signal mapping, real-time feedback, and performance analytics will then be developed and integrated. The complete system will undergo calibration, functional testing, therapist validation, and refinement. The initial deployment will target hospitals, physiotherapy clinics, rehabilitation centres, sports medicine centres, and medical colleges. Expected deliverables include a functional prototype, VR rehabilitation modules, therapist monitoring dashboard, patient performance data, and a validated system ready for further deployment and commercialization. The solution aims to improve patient engagement, therapy adherence, movement monitoring, and rehabilitation outcomes.

5. **Safety and Efficiency Analysis:**
   The system is designed to support safe and controlled upper-limb rehabilitation by allowing therapists to prescribe and monitor exercises according to the patient's

capability. The custom controller captures movement, grip, force, and Range of Motion (ROM), enabling controlled interaction with VR rehabilitation activities. Real-time visual and audio feedback helps the patient understand the required movement, while performance data allows therapists to identify progress and adjust therapy accordingly. The system reduces unnecessary repetition and improves therapy efficiency through interactive exercises, automated performance tracking, and objective progress measurement. Controlled testing, sensor calibration, therapist validation, and system optimization will be performed before clinical deployment.

6. **Scalability and Future Development:**
   The system is designed with modular hardware and software so that additional sensors, rehabilitation exercises, and software features can be integrated without replacing the complete system. Initial deployment can be expanded from hospitals, physiotherapy clinics, and rehabilitation centres to sports medicine centres, medical colleges, and research institutions. Future development can include home-based rehabilitation, tele-rehabilitation, AI-assisted personalized therapy, cloud-based progress monitoring, integration with Hospital Management Systems, and multi-language VR therapy modules. The software can be continuously upgraded while the reusable hardware remains in service, supporting long-term scalability and cost-effective deployment.

7. **Innovation Advantage:**
   The key innovation is the integration of a custom-built haptic rehabilitation controller with an immersive VR-based physiotherapy platform specifically designed for upper-limb motor rehabilitation. Unlike conventional physiotherapy, where exercises can become repetitive and progress is often assessed mainly through observation, the proposed system converts therapeutic movements into interactive VR activities while capturing movement, grip, force, and ROM data. The system provides real-time patient feedback and objective performance information to the therapist through a monitoring interface. The combination of custom rehabilitation hardware, motion and force sensing, VR-based gamification, real-time data processing, and performance tracking in one platform provides an engaging and measurable rehabilitation approach. Its indigenous and modular design also provides potential for affordable and scalable deployment compared with expensive advanced rehabilitation systems.

8. **Visual Impact & User Journey:**

